

理论力学

26 春-期末

题目来源于考后回忆，答案由 AI 生成。任课教师：邵立晶。

1. (含辅助变量的拉格朗日量) 考察包含辅助变量 χ 的拉格朗日量

$$L(q, \dot{q}, \chi) = \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2 + a\chi\dot{q} + b\chi^2.$$

- (i) 写出辅助变量 χ 满足的 Euler-Lagrange 方程;
(ii) 求解 χ , 并代入得到等效拉格朗日量 $L_{\text{eff}}(q, \dot{q})$;
(iii) 分析当参数满足何条件时, 会出现“负质量效应”, 导致运动不稳定 (即“快子不稳定”).

解答:

(i) χ 的 Euler-Lagrange 方程为

$$\frac{\partial L}{\partial \chi} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\chi}} = 0.$$

计算偏导数:

$$\frac{\partial L}{\partial \chi} = a\dot{q} + 2b\chi, \quad \frac{\partial L}{\partial \dot{\chi}} = 0,$$

故 χ 满足代数方程 (非动力学变量)

$$a\dot{q} + 2b\chi = 0.$$

(ii) 由上式立得

$$\chi = -\frac{a}{2b} \dot{q}.$$

代入原拉格朗日量:

$$\begin{aligned} L_{\text{eff}} &= \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2 + a\left(-\frac{a}{2b}\dot{q}\right)\dot{q} + b\left(-\frac{a}{2b}\dot{q}\right)^2 \\ &= \frac{1}{2}m\dot{q}^2 - \frac{1}{2}kq^2 - \frac{a^2}{2b}\dot{q}^2 + \frac{a^2}{4b}\dot{q}^2 \\ &= \frac{1}{2}\left(m - \frac{a^2}{2b}\right)\dot{q}^2 - \frac{1}{2}kq^2. \end{aligned}$$

(iii) 等效动能项系数为 $m_{\text{eff}} = m - \frac{a^2}{2b}$. 当 $m_{\text{eff}} < 0$ 时出现“负质量效应”, 即

$$\frac{a^2}{2b} > m \implies a^2 > 2bm.$$

此时等效动能项变号, 体系能量无下界, 导致“快子不稳定”。物理上这对应着辅助场 χ 与 q 的耦合过强, 使得有效惯量为负。

2. (Hamilton 矢量场与生成函数) 写出定义在相空间上的函数 $G(q, p) = pq$ 对应的 Hamilton 矢量场, 写出其对应的无穷小正则变换, 以及其对应的第二类生成函数。

解答:

Hamilton 矢量场:

$$X_G = \frac{\partial G}{\partial p} \frac{\partial}{\partial q} - \frac{\partial G}{\partial q} \frac{\partial}{\partial p} = q \frac{\partial}{\partial q} - p \frac{\partial}{\partial p}.$$

无穷小正则变换: 设无穷小参数为 ε , 则

$$\delta q = \varepsilon \{q, G\} = \varepsilon \frac{\partial G}{\partial p} = \varepsilon q, \quad \delta p = \varepsilon \{p, G\} = -\varepsilon \frac{\partial G}{\partial q} = -\varepsilon p.$$

即 $(q, p) \rightarrow (q + \varepsilon q, p - \varepsilon p)$, 这是相空间的标度变换 (伸缩变换)。有限形式为

$$Q = e^\varepsilon q, \quad P = e^{-\varepsilon} p.$$

第二类生成函数: 对无穷小正则变换, 第二类生成函数的形式为

$$F_2(q, P) = qP + \varepsilon G(q, P).$$

代入 $G(q, P) = qP$ 得

$$F_2(q, P) = qP + \varepsilon qP = (1 + \varepsilon)qP.$$

相应地, 有限形式的生成函数为

$$F_2(q, P) = e^\varepsilon qP,$$

其中 ε 为变换参数。验证:

$$p = \frac{\partial F_2}{\partial q} = e^\varepsilon P, \quad Q = \frac{\partial F_2}{\partial P} = e^\varepsilon q.$$

3. (阻尼振子) 考察阻尼振子满足的拉格朗日量

$$L(q, \dot{q}, t) = e^{2\lambda t} \left(\frac{1}{2} m \dot{q}^2 - \frac{1}{2} m \omega^2 q^2 \right).$$

- (i) 写出体系的哈密顿量 $H(q, p, t)$;
- (ii) 计算在第二类生成函数 $F_2(q, P, t) = e^{\lambda t} qP$ 对应的正则变换下得到的哈密顿量 $K(Q, P, t)$;
- (iii) 取 $\lambda = \omega$, 使用 Hamilton-Jacobi 方程求解 $q(t)$ 和 $p(t)$ 。

解答:

(i) 正则动量:

$$p = \frac{\partial L}{\partial \dot{q}} = m e^{2\lambda t} \dot{q} \implies \dot{q} = \frac{p}{m} e^{-2\lambda t}.$$

哈密顿量:

$$\begin{aligned} H(q, p, t) &= p\dot{q} - L = p \cdot \frac{p}{m} e^{-2\lambda t} - e^{2\lambda t} \left(\frac{1}{2} m \frac{p^2}{m^2} e^{-4\lambda t} - \frac{1}{2} m \omega^2 q^2 \right) \\ &= \frac{p^2}{m} e^{-2\lambda t} - \frac{p^2}{2m} e^{-2\lambda t} + \frac{1}{2} m \omega^2 e^{2\lambda t} q^2 \\ &= \frac{p^2}{2m} e^{-2\lambda t} + \frac{1}{2} m \omega^2 e^{2\lambda t} q^2. \end{aligned}$$

(ii) 由 $F_2 = e^{\lambda t} qP$:

$$p = \frac{\partial F_2}{\partial q} = e^{\lambda t} P, \quad Q = \frac{\partial F_2}{\partial P} = e^{\lambda t} q.$$

因此

$$q = e^{-\lambda t} Q, \quad p = e^{\lambda t} P.$$

新哈密顿量:

$$K(Q, P, t) = H(q, p, t) + \frac{\partial F_2}{\partial t}.$$

计算 $\partial F_2 / \partial t = \lambda e^{\lambda t} qP = \lambda QP$ 。

将 q, p 用 Q, P 表达代入 H :

$$\begin{aligned} H &= \frac{(e^{\lambda t} P)^2}{2m} e^{-2\lambda t} + \frac{1}{2} m \omega^2 e^{2\lambda t} (e^{-\lambda t} Q)^2 \\ &= \frac{P^2}{2m} + \frac{1}{2} m \omega^2 Q^2. \end{aligned}$$

故

$$K(Q, P, t) = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 Q^2 + \lambda QP.$$

(iii) 取 $\lambda = \omega$, 则新哈密顿量为

$$K(Q, P) = \frac{P^2}{2m} + \frac{1}{2}m\omega^2 Q^2 + \omega QP.$$

K 不含时 (时间显无关), 故可令主函数 $S(Q, \alpha, t) = W(Q, \alpha) - \alpha t$, 其中 α 为新的守恒动量。Hamilton-Jacobi 方程为

$$\frac{\partial S}{\partial t} + K\left(Q, \frac{\partial S}{\partial Q}\right) = 0 \implies \frac{1}{2m}\left(\frac{\partial W}{\partial Q}\right)^2 + \frac{1}{2}m\omega^2 Q^2 + \omega Q \frac{\partial W}{\partial Q} = \alpha.$$

这是 $\partial W / \partial Q$ 的二次方程:

$$\left(\frac{\partial W}{\partial Q}\right)^2 + 2m\omega Q \frac{\partial W}{\partial Q} + m^2\omega^2 Q^2 - 2m\alpha = 0.$$

配方得

$$\left(\frac{\partial W}{\partial Q} + m\omega Q\right)^2 = 2m\alpha \implies \frac{\partial W}{\partial Q} = -m\omega Q \pm \sqrt{2m\alpha}.$$

积分 (取 + 号, - 号等价) 得

$$W(Q, \alpha) = -\frac{1}{2}m\omega Q^2 + \sqrt{2m\alpha} Q.$$

故主函数为

$$S(Q, \alpha, t) = -\frac{1}{2}m\omega Q^2 + \sqrt{2m\alpha} Q - \alpha t.$$

新的守恒坐标 β 为

$$\beta = \frac{\partial S}{\partial \alpha} = \frac{mQ}{\sqrt{2m\alpha}} - t = Q\sqrt{\frac{m}{2\alpha}} - t.$$

因此

$$Q = \sqrt{\frac{2\alpha}{m}}(t + \beta).$$

新哈密顿量为零 ($K' = 0$), α, β 均为常数。由 $P = \partial S / \partial Q$ 得

$$P = \frac{\partial S}{\partial Q} = -m\omega Q + \sqrt{2m\alpha} = \sqrt{2m\alpha}[1 - \omega(t + \beta)].$$

回到原变量 ($q = e^{-\omega t}Q$, $p = e^{\omega t}P$):

$$q(t) = e^{-\omega t} \sqrt{\frac{2\alpha}{m}}(t + \beta), \quad p(t) = \sqrt{2m\alpha} e^{\omega t} [1 - \omega(t + \beta)].$$

引入 $A = \sqrt{2\alpha/m}$, $B = \beta$, 则简写为

$$q(t) = e^{-\omega t} A(t + B), \quad p(t) = m e^{\omega t} A[1 - \omega(t + B)].$$

这正是 $\lambda = \omega$ (临界阻尼) 情形下的精确解。与原轨迹方程 $\ddot{Q} = 0 \rightarrow Q = C_1 t + C_2$ 所得结果完全一致。

4. (隆格楞次矢量) 隆格楞次矢量定义为

$$\vec{A} = \vec{r} \times \vec{J} - \alpha m \hat{r},$$

其中 $\vec{J} = \vec{r} \times \vec{p}$ 为角动量, $\hat{r} = \vec{r}/r$ 为径向单位矢量, α 为常参数。计算 $\{A^i, J^j\}$ (Poisson 括号)。

解答:

在直角坐标下, $J^j = \varepsilon^{jmn} r_m p_n$ 。首先将 A^i 用坐标和动量表出:

$$\begin{aligned} A^i &= \varepsilon^{ikl} r_k J^l - \alpha m \frac{r^i}{r} \\ &= \varepsilon^{ikl} \varepsilon^{lmn} r_k r_m p_n - \alpha m \frac{r^i}{r}. \end{aligned}$$

利用恒等式 $\varepsilon^{ikl} \varepsilon^{lmn} = \varepsilon^{lik} \varepsilon^{lmn} = \delta^{im} \delta^{kn} - \delta^{in} \delta^{km}$:

$$\varepsilon^{ikl} \varepsilon^{lmn} r_k r_m p_n = (\delta^{im} \delta^{kn} - \delta^{in} \delta^{km}) r_k r_m p_n = r_n r^i p_n - r_m r_m p^i = r^i (\vec{r} \cdot \vec{p}) - r^2 p^i.$$

因此

$$A^i = r^i (\vec{r} \cdot \vec{p}) - r^2 p^i - \alpha m \frac{r^i}{r}.$$

现在计算 Poisson 括号 $\{A^i, J^j\}$ 。利用 J^j 是旋转生成元这一事实, $\{A^i, J^j\}$ 应当体现矢量 $\vec{A}$ 的旋转性质:

$$\boxed{\{A^i, J^j\} = \varepsilon^{ijk} A^k.}$$

这一结果可直接验证。以 $j = 1$ 为例 ($J^1 = yp_z - zp_y$):

计算 $\{A^2, J^1\} \stackrel{?}{=} \varepsilon^{213} A^3 = -A^3$ 。由 $\{r^i, J^j\} = \varepsilon^{ijk} r^k$, $\{p^i, J^j\} = \varepsilon^{ijk} p^k$, 以及 A^i 是 r^i, p^i 的矢量组合 (A^i 中各项均是矢量的分量), 根据 Poisson 括号的链式法则可得上述结果。

验证要点:

- $\{r^i (\vec{r} \cdot \vec{p}), J^j\} = \{r^i, J^j\} (\vec{r} \cdot \vec{p}) + r^i \{\vec{r} \cdot \vec{p}, J^j\} = \varepsilon^{ijk} r^k (\vec{r} \cdot \vec{p}) + r^i \cdot 0$ (因 $\vec{r} \cdot \vec{p}$ 是转动标量);
- $\{r^2 p^i, J^j\} = \{r^2, J^j\} p^i + r^2 \{p^i, J^j\} = 0 \cdot p^i + r^2 \varepsilon^{ijk} p^k = \varepsilon^{ijk} r^2 p^k$;
- $\{\frac{r^i}{r}, J^j\} = \frac{1}{r} \{r^i, J^j\} + r^i \{\frac{1}{r}, J^j\} = \frac{1}{r} \varepsilon^{ijk} r^k + 0$ 。

将各部分组合即得

$$\{A^i, J^j\} = \varepsilon^{ijk} [r^k (\vec{r} \cdot \vec{p}) - r^2 p^k - \alpha m \frac{r^k}{r}] = \varepsilon^{ijk} A^k.$$

这正表明 $\vec{A}$ 在空间旋转下如同一个三维矢量变换, 它是 $\vec{r}$ 和 $\vec{p}$ 构成的唯一与 $\vec{J}$ 对合的矢量守恒量 (在 Kepler 问题中)。

5. (场论——哈密顿原理与正则方程)

- 一个 3+1 维体系 (t, x, y, z) 的拉氏量密度为 $\mathcal{L}(\eta, \eta_{,\mu}, \eta_{,\mu\nu}, x^\mu)$, 使用哈密顿原理给出 η 满足的场方程;
- 一个 1+1 维体系 (t, x) 的哈密顿量密度为 $\mathcal{H}(\eta, \eta_x, \pi, \pi_x, \pi_{xx})$, 给出其对应的正则方程组。

解答:

(i) 作用量为 $S[\eta] = \int d^4x \mathcal{L}(\eta, \eta_{,\mu}, \eta_{,\mu\nu}, x^\mu)$ 。哈密顿原理要求 $\delta S = 0$:

$$\begin{aligned} \delta S &= \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \eta} \delta \eta + \frac{\partial \mathcal{L}}{\partial \eta_{,\mu}} \delta \eta_{,\mu} + \frac{\partial \mathcal{L}}{\partial \eta_{,\mu\nu}} \delta \eta_{,\mu\nu} \right] \\ &= \int d^4x \frac{\partial \mathcal{L}}{\partial \eta} \delta \eta - \int d^4x \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \eta_{,\mu}} \right) \delta \eta + \int d^4x \partial_\mu \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial \eta_{,\mu\nu}} \right) \delta \eta, \end{aligned}$$

其中用到了两次分部积分, 并设边界上 $\delta \eta$ 及其一阶导数为零。

令 $\delta S = 0$ 对任意 $\delta \eta$ 成立, 得到

$$\frac{\partial \mathcal{L}}{\partial \eta} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial \eta_{,\mu}} \right) + \partial_\mu \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial \eta_{,\mu\nu}} \right) = 0.$$

这是含二阶导数的广义 Euler-Lagrange 方程。

(ii) 正则变量为 η 和 π , 哈密顿量为 $H = \int dx \mathcal{H}(\eta, \eta_{,x}, \pi, \pi_{,x}, \pi_{,xx})$ 。
由变分原理 $\delta \int dt dx (\pi \dot{\eta} - \mathcal{H}) = 0$, 得正则方程:

$$\dot{\eta} = \frac{\delta H}{\delta \pi}, \quad \dot{\pi} = -\frac{\delta H}{\delta \eta},$$

其中 $\delta/\delta\pi$ 表示泛函导数。

对 $\mathcal{H}$ 含 π 的二阶空间导数情形:

$$\begin{aligned} \dot{\eta} &= \frac{\partial \mathcal{H}}{\partial \pi} - \partial_x \left(\frac{\partial \mathcal{H}}{\partial \pi_{,x}} \right) + \partial_x^2 \left(\frac{\partial \mathcal{H}}{\partial \pi_{,xx}} \right), \\ \dot{\pi} &= -\frac{\partial \mathcal{H}}{\partial \eta} + \partial_x \left(\frac{\partial \mathcal{H}}{\partial \eta_{,x}} \right). \end{aligned}$$

泛函导数由 $\frac{\delta H}{\delta \pi} = \sum_{n=0}^{\infty} (-1)^n \partial_x^n \frac{\partial \mathcal{H}}{\partial (\partial_x^n \pi)}$ 给出。

6. (复标量电子场) 复标量电子场的拉格朗日量密度为

$$\mathcal{L}_0 = \phi^\mu \phi_{,\mu}^* - m^2 \phi \phi^*,$$

其对应的 U(1) Noether 流为 $j^\mu = i(\phi^* \phi^{,\mu} - \phi \phi^{*,\mu})$ 。在外场 A_μ 下, 拉格朗日量密度会多一项 $j^\mu A_\mu$, 即 $\mathcal{L} = \mathcal{L}_0 + j^\mu A_\mu$ 。

- (i) 给出 ϕ, ϕ^* 满足的场方程;
- (ii) 给出守恒流与对应的守恒定律。

解答:

(i) 总拉格朗日量密度:

$$\mathcal{L} = \partial_\mu \phi \partial^\mu \phi^* - m^2 \phi \phi^* + i A_\mu (\phi^* \partial^\mu \phi - \phi \partial^\mu \phi^*).$$

注意第二项中 A_μ 与 $\phi^* \partial^\mu \phi$ 均为普通的数 (经典场), 可交换顺序。

对 ϕ^* 做变分:

$$\frac{\partial \mathcal{L}}{\partial \phi^*} = -m^2 \phi + i A_\mu \partial^\mu \phi = -m^2 \phi + i A^\mu \partial_\mu \phi,$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^*)} = \partial^\mu \phi - i A^\mu \phi.$$

代入 Euler-Lagrange 方程 $\frac{\partial \mathcal{L}}{\partial \phi^*} - \partial_\mu \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^*)} = 0$:

$$\begin{aligned} 0 &= -m^2 \phi + i A^\mu \partial_\mu \phi - \partial_\mu (\partial^\mu \phi - i A^\mu \phi) \\ &= -m^2 \phi + i A^\mu \partial_\mu \phi - \square \phi + i \partial_\mu (A^\mu \phi) \\ &= -\square \phi - m^2 \phi + i A^\mu \partial_\mu \phi + i (\partial_\mu A^\mu) \phi + i A^\mu \partial_\mu \phi \\ &= -\square \phi - m^2 \phi + 2i A^\mu \partial_\mu \phi + i (\partial_\mu A^\mu) \phi. \end{aligned}$$

整理得 ϕ 的场方程

$$\square \phi + m^2 \phi - 2i A^\mu \partial_\mu \phi - i (\partial_\mu A^\mu) \phi = 0.$$

对 ϕ 做变分类似可得 ϕ^* 的共轭场方程：

$$\square\phi^* + m^2\phi^* + 2iA^\mu\partial_\mu\phi^* + i(\partial_\mu A^\mu)\phi^* = 0.$$

注：若希望将方程写成协变导数的紧凑形式，注意到 $(\partial_\mu - iA_\mu)^2\phi + m^2\phi = \square\phi - 2iA^\mu\partial_\mu\phi - i(\partial_\mu A^\mu)\phi - A^2\phi + m^2\phi$ 。由于拉氏量只加了 $j^\mu A_\mu$ 项而未包含 A^2 项，场方程中不含 $A^2\phi$ ，因此不等于通常的 $D_\mu D^\mu\phi + m^2\phi = 0$ 。这正是“外场”处理的含义—— A_μ 是给定的背景场，不参与动力学。

(ii) 总拉格朗日量 $\mathcal{L} = \mathcal{L}_0 + j^\mu A_\mu$ 仍具有整体 U(1) 对称性 (A_μ 作为背景场不参与变换)：

$$\phi \rightarrow e^{-i\alpha}\phi, \quad \phi^* \rightarrow e^{+i\alpha}\phi^*,$$

无穷小形式为 $\delta\phi = -i\alpha\phi$, $\delta\phi^* = +i\alpha\phi^*$ 。

由 Noether 定理，守恒流需对 $\mathcal{L}$ 而非仅 $\mathcal{L}_0$ 计算：

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \frac{\delta\phi}{\alpha} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)} \frac{\delta\phi^*}{\alpha}.$$

偏导数为

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \partial^\mu\phi^* + iA^\mu\phi^*, \quad \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)} = \partial^\mu\phi - iA^\mu\phi.$$

代入得

$$\begin{aligned} J^\mu &= (\partial^\mu\phi^* + iA^\mu\phi^*)(-i\phi) + (\partial^\mu\phi - iA^\mu\phi)(+i\phi^*) \\ &= -i\phi\partial^\mu\phi^* + A^\mu|\phi|^2 + i\phi^*\partial^\mu\phi + A^\mu|\phi|^2 \\ &= i(\phi^*\partial^\mu\phi - \phi\partial^\mu\phi^*) + 2A^\mu|\phi|^2. \end{aligned}$$

因此含外场的总守恒流为

$$\boxed{J^\mu = j^\mu + 2A^\mu|\phi|^2, \quad j^\mu \equiv i(\phi^*\partial^\mu\phi - \phi\partial^\mu\phi^*)}.$$

守恒律验证：由第 (i) 问得到的场方程

$$\begin{aligned} \square\phi + m^2\phi - 2iA^\mu\partial_\mu\phi - i(\partial_\mu A^\mu)\phi &= 0, \\ \square\phi^* + m^2\phi^* + 2iA^\mu\partial_\mu\phi^* + i(\partial_\mu A^\mu)\phi^* &= 0, \end{aligned}$$

计算

$$\begin{aligned} \partial_\mu J^\mu &= i(\partial_\mu\phi^*\partial^\mu\phi + \phi^*\square\phi - \partial_\mu\phi\partial^\mu\phi^* - \phi\square\phi^*) + 2\partial_\mu(A^\mu|\phi|^2) \\ &= i(\phi^*\square\phi - \phi\square\phi^*) + 2(\partial_\mu A^\mu)|\phi|^2 + 2A^\mu\partial_\mu|\phi|^2. \end{aligned}$$

代入场方程，并利用 $\partial_\mu|\phi|^2 = \phi^*\partial_\mu\phi + \phi\partial_\mu\phi^*$ ，各项相互抵消，故

$$\boxed{\partial_\mu J^\mu = 0}.$$

对应的守恒荷为

$$Q = \int d^3x J^0 = \int d^3x [i(\phi^*\dot{\phi} - \dot{\phi}\phi^*) + 2A^0|\phi|^2], \quad \frac{dQ}{dt} = 0.$$

物理意义：外场 A_μ 存在时，U(1) 对称性依然保持，但守恒流 J^μ 相比自由情形多出一项 $2A^\mu|\phi|^2$ ，反映外场对荷密度分布的贡献。